

Blind Disturbance-Model Selection for Diffusion-Aided Semantic Image Reception

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Abstract—We present a blind disturbance-model selector for diffusion-aided semantic image reception under unknown co-channel interference. Received energy and block-energy heterogeneity determine whether Gaussian denoising or joint signal-interference reconstruction recovers the desired latent. A common decoder then produces the image. Selection precedes diffusion inference and executes one branch, so each frame incurs the cost of its chosen recovery model. The rule uses $O(n)$ calculations, with thresholds fitted on development images and frozen neural weights. Experiments use CelebA images with CIFAR-10 interference and identical observations across receivers. On 256 independent image pairs with two random seeds and 18 channel conditions, the frozen rule reaches 21.593 dB mean PSNR. It improves on calibrated joint sampling by +1.334 dB and on blind Gaussian denoising by +0.750 dB. The rule uses 52.2 predictor evaluations per frame on average, compared with 80 for joint sampling. The same-gate comparison isolates the gain beyond bypassing joint inference. Separately trained DeepJSCC and MambaJSCC codecs provide external references under the same channel-use budget.

Index Terms—Semantic communications, interference cancellation, diffusion models, receiver adaptation, deep joint source-channel coding.

I. INTRODUCTION

Semantic image communication requires recovering image content from limited, corrupted channel observations. Deep joint source-channel coding (JSCC) addresses this task by mapping images directly to channel symbols, with gradual quality degradation as noise increases [1]. SwinJSCC and MambaJSCC extend this approach through transformer and state-space architectures [2], [3]. Generative receivers further exploit learned priors to recover corrupted representations. Even so, co-channel interference of unknown power and structure leaves a deployed receiver with a choice: approximate the disturbance as noise or reconstruct the interfering signal explicitly.

CDDM and ICDM provide two relevant recovery mechanisms. CDDM matches the aggregate disturbance to a diffusion noise level [4], whereas ICDM jointly reconstructs desired and interfering latents [5]. Iterative source error correction [6], channel-conditioned diffusion reconstruction [7], and blind inverse-problem inference [8] also advance receiver adaptation. However, joint reconstruction must infer two latents and unknown interference amplitudes from one mixture. Estimation errors can offset the benefit of modeling interference structure while consuming additional computation. Gaussian denoising avoids that estimation burden but discards useful structure. We ask whether received symbols can guide this

model choice before either generative receiver reconstructs the image.

We propose a blind disturbance-model selector that takes a received latent and chooses either Gaussian denoising or joint reconstruction. The selected receiver recovers the latent for a common image decoder. The selector computes two energy statistics in $O(n)$ time and executes only one branch. Holding the encoders, priors, and decoder fixed isolates the effect of receiver choice. Selection leaves the transmitted representation unchanged and adds no predictor evaluations.

Explicit interference modeling offers structure at the cost of estimation uncertainty, so its value can vary across frames. We use excess received energy to measure aggregate disturbance and block-energy heterogeneity to describe its variation across latent blocks. Desired-signal variation affects both statistics, so we determine their decision value empirically. We use image-group validation to compare energy-only and two-feature rules on separate development images, retaining a fixed receiver if adaptation gives no validation gain. The chosen rule is frozen before independent confirmation.

Experiments on 256 independent image pairs show a mean PSNR gain of 0.750 dB over blind Gaussian denoising and 1.334 dB over calibrated joint sampling, with 34.7% fewer predictor evaluations than joint sampling. A same-gate control separates the change of recovery model from bypassing joint inference. The selector is fitted once and processes every confirmation frame using its received statistics. Profile-wise comparisons reveal that the preferred fixed receiver changes with interference structure, while harmful decisions and block mismatch delimit the benefit. Separately trained DeepJSCC and MambaJSCC place the receiver results alongside external codec references. The main comparison uses a common SwinJSCC latent space to attribute quality and computation differences to the receiver decision.

Our contributions are threefold. (1) We introduce a blind selector that chooses a disturbance model before generative inference using two energy statistics. (2) We show that the preferred fixed receiver differs across interference profiles, motivating observation-dependent selection. (3) We establish a mean-quality and computation benefit on independent paired observations, while quantifying harmful decisions and mismatch sensitivity. Section II presents the method, Section III evaluates it, and Section IV concludes the paper.

II. METHOD

In this section, we develop a blind selector that maps a received latent to one of two generative receivers. We first define the communication model, recovery models, and selection objective, then describe the candidate receivers and the rule fitted on development images.

A. Communication Model

Let $\mathbf{u} \in [0, 1]^d$ be an RGB image, with $d = 3H_{\text{img}}W_{\text{img}}$. The JSCC encoder f_s combines representation learning and channel mapping. f_z encodes an interfering image $\mathbf{v}$ into the same dimension:

$$\mathbf{x} = \mathcal{N}(f_s(\mathbf{u})), \quad \mathbf{z} = \mathcal{N}(f_z(\mathbf{v})), \quad \mathcal{N}(\mathbf{t}) = \frac{\sqrt{n}\mathbf{t}}{\|\mathbf{t}\|}. \quad (1)$$

Each nonzero codeword has unit average power and bandwidth ratio n/d . For B blocks of lengths L_b , with $\sum_b L_b = n$, the received symbols are

$$\mathbf{y}_b = \mathbf{x}_b + a_b \mathbf{z}_b + \mathbf{w}_b, \quad \mathbf{w} \sim \mathcal{CN}(\mathbf{0}, N_0 \mathbf{I}), \quad a_b \geq 0. \quad (2)$$

Here a_b is the unknown interference amplitude and N_0 the complex noise power. The desired-channel gain is unity and symbol alignment is known. The receiver knows N_0 , the normalization, and block boundaries, but not $\mathbf{u}$, $\mathbf{z}$, or a_b . For nominal SINR γ , the simulator sets the average interference power to $10^{-\gamma/10} - N_0$. Realized block energies of $\mathbf{z}$ make the per-frame SINR differ from this nominal setting.

B. Generative Image Recovery

The receiver combines the frame-specific evidence in $\mathbf{y}$ with learned latent priors. Joint reconstruction uses both priors:

$$p(\mathbf{x}, \mathbf{z} \mid \mathbf{y}, \mathbf{a}) \propto p_s(\mathbf{x})p_z(\mathbf{z}) \times \exp\left[-\frac{1}{N_0} \sum_b \|\mathbf{y}_b - \mathbf{x}_b - a_b \mathbf{z}_b\|^2\right]. \quad (3)$$

The amplitudes require estimation. The observation likelihood alone admits $(\mathbf{x}_b + a_b \delta_b, \mathbf{z}_b - \delta_b)$ with the same mixture, leaving the priors to resolve the ambiguity. Gaussian recovery uses p_s with an aggregate disturbance likelihood:

$$p_G(\mathbf{y} \mid \mathbf{x}; \nu) = \mathcal{CN}(\mathbf{y}; \mathbf{x}, \nu \mathbf{I}), \quad \nu \geq N_0. \quad (4)$$

The Gaussian and joint receivers, $\mathcal{R}_G$ and $\mathcal{R}_S$, use approximate diffusion recovery with observation-based initialization or guidance. Joint recovery also estimates the interfering latent. A blind rule $\pi_\theta(\mathbf{y}) \in \{G, S\}$ selects one branch. The common decoder then reconstructs

$$\hat{\mathbf{u}}_\theta = g_s(\mathcal{N}(\mathcal{R}_{\pi_\theta(\mathbf{y})}(\mathbf{y}))). \quad (5)$$

C. Receiver-Selection Objective

We measure image reconstruction with per-image PSNR, $q(\mathbf{u}, \hat{\mathbf{u}}) = -10 \log_{10}(\|\mathbf{u} - \hat{\mathbf{u}}\|^2/d)$ for clipped RGB outputs. LPIPS and MS-SSIM complement PSNR with perceptual and structural fidelity. Computation counts calls to both priors. Receiver latency includes selection and decoding.

For development pairs $\mathcal{D} = \{(\mathbf{u}_i, \mathbf{y}_i)\}_{i=1}^M$, we fit

$$\theta^* \in \arg \max_{\theta \in \Theta} \frac{1}{M} \sum_{i=1}^M q(\mathbf{u}_i, \hat{\mathbf{u}}_\theta(\mathbf{y}_i)), \quad (6)$$

where Θ contains the rule family retained by image-group validation and a constant-receiver fallback. Bandwidth, power, and neural weights remain fixed, with computation reported separately. After offline fitting, the frozen rule executes one branch without knowing its reconstruction error.

D. Candidate Receivers

Both receivers share the frozen desired-signal prior and decoder, isolating the disturbance-model choice. To account for unknown interference power, the Gaussian receiver estimates aggregate disturbance from the received symbols:

$$e_b = \frac{\|\mathbf{y}_b\|^2}{L_b}, \quad \bar{e} = \frac{1}{n} \sum_b L_b e_b, \quad (7)$$

$$\hat{P} = [\bar{e} - 1 - N_0]_+, \quad \hat{\nu} = N_0 + \hat{P}.$$

With unit source power and a vanishing signal–interference cross expectation, unclipped excess energy estimates the mean interference energy. Finite blocks, latent correlation, and clipping make this estimate inexact. In particular, local desired-signal energy can resemble interference heterogeneity.

Let $r(\mathbf{y}) \in \mathbb{R}^{2n}$ stack real and imaginary components. Given the training diffusion schedule $\bar{\alpha}_t$, the Gaussian receiver G starts at the nearest noise level satisfying

$$\frac{1 - \bar{\alpha}_{t_0}}{\bar{\alpha}_{t_0}} \simeq \frac{\hat{\nu}}{2}, \quad \mathbf{v}_{t_0} = \sqrt{\bar{\alpha}_{t_0}} r(\mathbf{y}). \quad (8)$$

The factor 1/2 converts complex noise power to variance per real coordinate. For decreasing respaced indices, a noise predictor ϵ_s gives

$$\hat{\mathbf{x}}_0 = \frac{\mathbf{v}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_s(\mathbf{v}_t, t)}{\sqrt{\bar{\alpha}_t}}, \quad (9)$$

$$\mathbf{v}_{t'} = \sqrt{\bar{\alpha}_{t'}} \hat{\mathbf{x}}_0 + \sqrt{1 - \bar{\alpha}_{t'}} \epsilon_s(\mathbf{v}_t, t). \quad (10)$$

The receiver uses at most 40 predictor evaluations and fewer when the matched initial index is below 39.

To exploit interference structure, the joint receiver S initializes $a_b^{(0)} = \sqrt{[e_b - 1 - N_0]_+}$ and performs joint diffusion inference with signal and interference predictors. During late corrector updates it obtains denoised, frame-normalized estimates $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{z}}$, then updates

$$a_b^+ = \left[\frac{\Re\{\tilde{\mathbf{z}}_b^H(\mathbf{y}_b - \tilde{\mathbf{x}}_b)\}}{\|\tilde{\mathbf{z}}_b\|^2} \right]_+. \quad (11)$$

An empty denominator retains the previous amplitude. This nonnegative least-squares update uses point estimates, leaving their uncertainty outside the amplitude fit. Calibration begins when the signal coefficient $\alpha(t) \geq 0.5$. Estimated block SINR updates the inherited guidance coefficients by clipped linear interpolation of the original AWGN table. The existing 40-step, order-three sampler calls each prior 40 times. An initialization-only variant keeps the amplitudes fixed and isolates the effect of calibration.

E. Blind Selection Before Inference

To distinguish frames with similar total energy but different block profiles, the selector combines $\hat{P}$ with normalized block-energy variation:

$$H = \frac{B^{-1} \sum_{b=1}^B (e_b - \bar{e})^2}{\max(\bar{e}^2, \varepsilon)}. \quad (12)$$

Here $\varepsilon > 0$ prevents division by zero, and all receiver blocks have equal length. The statistic H captures variation in the mixture's block energies. For $\theta = (\tau_\ell, \tau_h, \eta)$, the rule is

$$\pi_\theta(\mathbf{y}) = \begin{cases} \text{G}, & \hat{P} \leq \tau_\ell, H \leq \eta, \\ \text{G}, & \hat{P} \leq \tau_h, H > \eta, \\ \text{S}, & \text{otherwise.} \end{cases} \quad (13)$$

We test both assignment directions of G and S in (13), learning which side of the threshold uses Gaussian recovery. The energy-only family sets $\tau_\ell = \tau_h$. Either family can reduce to a constant receiver, allowing validation to reject adaptation.

To assess transfer across images, eight development folds keep each image's conditions together. We fit thresholds on seven folds and evaluate PSNR on the eighth, then refit the best out-of-fold family on all development images. Ties favor lower cost, then the simpler family. If neither family improves on the better fixed receiver, validation retains that receiver. We freeze the thresholds before confirmation and pressure testing. At inference, the selector evaluates (13) and executes only the chosen branch. This adds $O(n)$ energy calculations and no diffusion evaluations.

For analysis, let $q_j(\mathbf{u}, \mathbf{y})$ be a candidate's PSNR. The unavailable per-frame oracle has quality $\max(q_G, q_S)$. The exact identity

$$\max(q_G, q_S) - q_{\pi_\theta} = |q_G - q_S| \mathbf{1}\{\pi_\theta \neq \arg \max_j q_j\} \quad (14)$$

expresses regret as the quality gap on frames assigned to the worse receiver. Large-gap errors therefore dominate mean regret. We optimize mean PSNR and evaluate MSE, LPIPS, and harmful decisions separately.

III. EXPERIMENTS

In this section, we test whether selecting one frozen receiver improves image quality while reducing computation. We compare fixed receivers and external codecs, then examine harmful decisions and interference mismatch.

A. Experimental Setup

We use 128×128 CelebA center crops as desired images [9], while 32×32 CIFAR-10 interference images are nearest-neighbor upsampled. Training uses 162,770 desired and 50,000 interference images. The frozen SwinJSCC codec has stage widths (128, 192, 256), depths (2, 2, 6), and an $8 \times 16 \times 16$ real latent: 1,024 complex symbols at CBR 1/48. Each DiT prior has 28 blocks, width 1,152, 16 heads and patch size two. The inherited training schedule uses 40/100 desired/interference-codec epochs and 50/20

desired/interference-prior epochs. Selection adds no neural training.

DeepJSCC and MambaJSCC each train desired and interference codecs for 40 epochs at SNR 20 dB and CBR 1/48, with one seed and validation-MSE checkpoint selection. DeepJSCC uses a pinned third-party PyTorch reproduction, and MambaJSCC uses official code. Public checkpoints also provide transfer diagnostics without fine-tuning, using one encoder for both streams. Public MambaJSCC differs in training domain, channel and resolution; public DeepJSCC also uses CBR 0.16256. These direct codecs provide external references with their own latent spaces. The Gaussian and joint comparisons are receiver-only ports sharing frozen SwinJSCC weights and DiT priors. A full CDDM/ICDM comparison would require each system's codec training and matching diffusion priors.

Development uses 64 validation pairs and seed 1700. For independent confirmation, we use 256 test pairs, seeds 2700/2701, SNR 20 dB, and nominal SINRs $\{-7, -4, 0, 4, 7, 10\}$ dB. Sixteen 64-symbol blocks have stationary, alternating (0.25, 1.75), or burst (0, 2) relative interference powers. These 18 conditions give 9,216 frames per receiver. We run confirmation only after freezing the policy. File hashes exclude overlap with development, pressure tests, and 200 previously evaluated pairs. Splits are independent at image-pair level, with identities unchecked. The power grid is $\{0, 0.1, 0.2, 0.35, 0.5, 0.75, 1, 1.5, 2, 3, 4, 8, 10^6\}$, and the heterogeneity grid is $\{0.05, 0.1, 0.2, 0.4, 0.8, 1.6\}$, giving 26 energy-only and 2,028 two-feature rules across both assignment directions.

We compare with direct decoding, Gaussian reception using N_0 alone, blind Gaussian reception, initialization-only and calibrated joint sampling, and the selector. The same-gate control uses direct decoding for $\hat{P} \leq 0.15$ and calibrated joint sampling otherwise. Its outputs and component times are reconstructed from paired branches, excluding gate overhead. Development also tests coupled feedback and a Gaussian receiver given nominal disturbance power, a privileged diagnostic. All SwinJSCC variants share received tensors and sampler seeds. External codecs share image pairs, conditions and channel seeds but generate their own codewords.

Measurements use float32, batch size one, TF32 disabled, four CPU threads, and an RTX 4090. Synchronized receiver time includes selection, sampling and decoding, excluding encoding, metrics and loading. Function evaluations (NFE) count calls to both priors. Quality metrics are per-image PSNR and MSE, LPIPS-VGG v0.1 [10], and four-scale MS-SSIM [11], averaged arithmetically after RGB clipping to $[0, 1]$. Four scales fit the image resolution. A harmful decision loses more than 0.5 dB against its comparator.

B. Frozen Rule and Receiver Comparison

We first test whether block heterogeneity improves rule selection under the same image-group folds. Energy-only selection yields 22.002 dB, whereas the two-feature family yields 22.239 dB. We therefore retain the energy and hetero-

TABLE I
RECONSTRUCTION QUALITY AND RECEIVER COST ON THE
CONFIRMATION SET.

Receiver	PSNR	MSE $\times 10^3$	LPIPS	MS-SSIM	NFE	ms
SwinJSCC direct	18.67	29.14	0.440	0.697	0.0	8
DeepJSCC direct	17.39	24.71	0.575	0.684	0.0	1
MambaJSCC direct	16.92	44.13	0.463	0.613	0.0	10
Gaussian (N0)	18.83	28.70	0.438	0.700	18.0	225
Blind Gaussian	20.84	24.90	0.378	0.782	40.0	488
Init. joint	19.61	20.15	0.363	0.793	80.0	1110
Calibrated joint	20.26	18.49	0.362	0.802	80.0	1180
Joint + gate	20.45	18.43	0.361	0.801	65.9	974
Selector	21.59	16.45	0.336	0.832	52.2	701

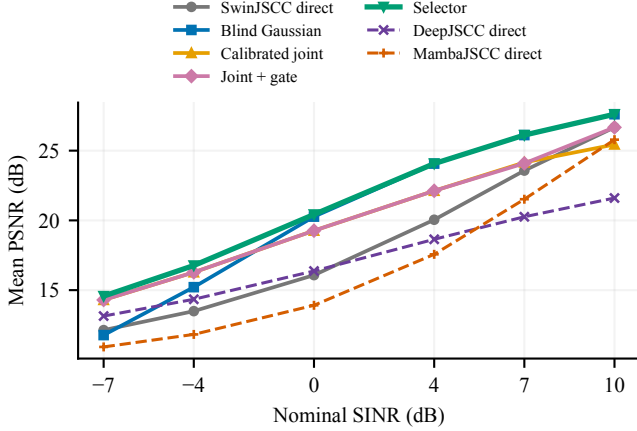


Fig. 1. Mean PSNR versus nominal SINR over the three interference profiles.

generality rule with $\tau_\ell = 8$, $\tau_h = 0.5$ and $\eta = 0.2$, with Gaussian reception below the applicable power threshold.

The low-heterogeneity development region contains 810 frames and has a maximum excess-energy estimate of 5.241. The fitted threshold 8 therefore lies beyond observed support and makes this region always Gaussian. The transition beyond this region remains unobserved, and family-level validation leaves each feature’s individual contribution unresolved. The earlier coupled-feedback receiver obtains 20.477 dB, versus 20.605 dB for calibrated joint sampling and 20.835 dB for its same-gate control. The privileged nominal-power Gaussian diagnostic obtains 21.591 dB. These development results informed the receiver design before the independent evaluation. We next test the frozen rule on independent paired observations (Table I and Fig. 1). The selector changes mean PSNR by +1.334 dB relative to calibrated joint sampling, +0.750 dB relative to blind Gaussian denoising, and +1.140 dB relative to the same-gate control. The gain over the same-gate control distinguishes changing the recovery model from the tested bypass alone. Estimating total disturbance raises the Gaussian receiver from 18.829 to 20.843 dB, showing the value of accounting for interference power.

To examine where selection helps, we compare fixed receivers across profiles. Under stationary interference, Gaussian denoising reaches 21.310 dB, compared with 19.282 dB

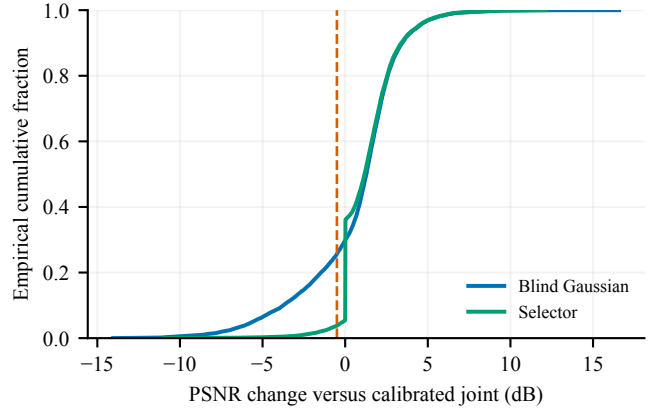


Fig. 2. Empirical CDF of per-frame PSNR change relative to calibrated joint sampling.

for joint sampling. Under burst interference, joint sampling reaches 21.317 dB, compared with 20.481 dB for Gaussian denoising. The selector reaches 21.317, 21.285, and 22.177 dB for stationary, alternating, and burst interference, respectively. This reversal creates an opportunity for observation-dependent receiver choice.

Matched direct-codec references reach 17.394 dB for DeepJSCC and 16.923 dB for MambaJSCC in their independently trained latent spaces. These results describe the single-seed, 40-epoch training budget; convergence and architecture-wide rankings remain untested. Without fine-tuning, public DeepJSCC (ImageNet/SNR 19), MambaJSCC (CLIC2021/AWGN 10), and MambaJSCC (DIV2K/Rayleigh) weights yield 13.755, 17.020, and 16.650 dB, respectively. The training and protocol differences make these values transfer diagnostics. Figure 1 uses solid curves for the frozen SwinJSCC latent space and dashed curves for the separately trained codecs.

C. Inference Cost and Harmful Decisions

Figure 2 shows the frame-level CDF, with a vertical line marking the -0.5 dB harmful-loss threshold. The selector uses the less expensive Gaussian branch on 69.5% of confirmation frames. It averages 52.2 predictor evaluations and 701 ms, versus 80 evaluations and 1180 ms for joint sampling. Against joint sampling, 3.79% of frames lose more than 0.5 dB, and the fifth percentile of the paired change is -0.127 dB (Fig. 2). The unavailable per-frame oracle reaches 21.760 dB; selection leaves 0.167 dB mean oracle regret. Mean MSE changes from 0.01849 to 0.01645, LPIPS from 0.3618 to 0.3362, and MS-SSIM from 0.8015 to 0.8318. The mean fidelity improvements coexist with the losses visible in the lower CDF tail.

D. Block Mismatch and No Interference

To test block-partition sensitivity, we use 64 additional pairs, seed 3700, and SINR $-4, 4, 10$ dB. One profile shifts burst boundaries by half a receiver block. Another draws segment lengths uniformly from 16 to 128 symbols and normalizes the

TABLE II
MEAN PSNR UNDER BLOCK MISMATCH AND NO INTERFERENCE.

Profile	Direct	Gaussian	Joint	Select
Shifted blocks	20.83	22.99	21.09	23.00
Random lengths	20.64	23.42	22.74	23.98
No interference	31.37	31.37	27.73	31.37

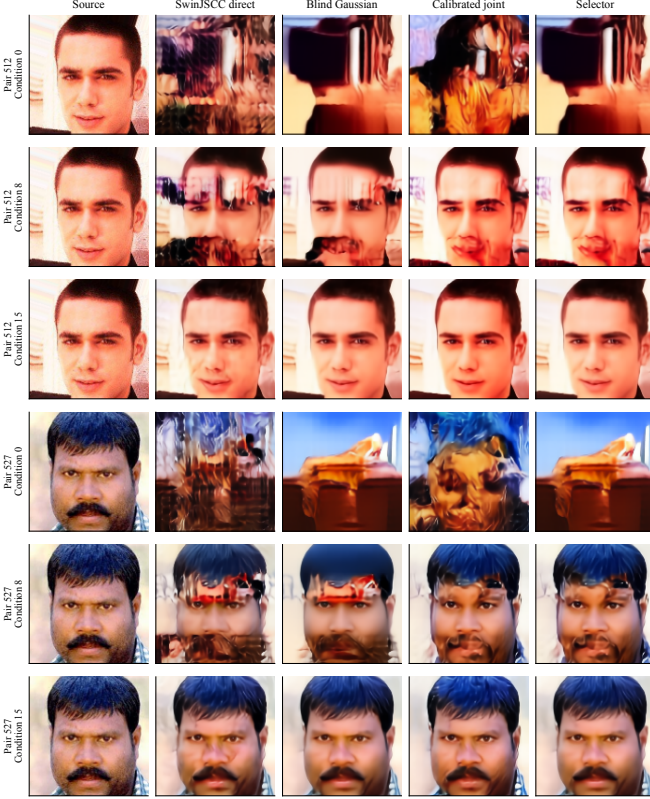


Fig. 3. Reconstructed images for six prespecified confirmation examples.

binary on/off amplitude pattern. A separate SNR-20 condition has no interference. The receiver keeps its original block partition and frozen rule. We report these diagnostic families separately from matched confirmation. Table II reports the robustness tests. The selector changes PSNR relative to Gaussian reception by +0.010 dB for shifted blocks and +0.563 dB for random lengths. With no interference, it matches Gaussian reception and avoids the joint receiver’s loss. Shifting the burst boundaries reduces the selection benefit to almost zero, identifying sensitivity to the receiver’s block partition. For a qualitative check, Fig. 3 shows all six prespecified confirmation examples: image pairs 512 and 527 under conditions 0, 8, and 15. No example was selected by reconstruction quality.

IV. DISCUSSION AND CONCLUSION

In this section, we relate receiver complementarity to the selection gain and its sensitivity to interference structure. Gaussian denoising leads under stationary interference, whereas

joint reconstruction leads under burst interference. Energy-based selection exploits this change of ordering without running both receivers. Its mean gain coexists with harmful frame decisions, and shifting the burst boundaries nearly removes the gain over Gaussian denoising. The decision therefore depends on how received energy is distributed across the receiver’s blocks.

In this paper, we presented a blind selector that uses received energy and block-energy heterogeneity to choose between Gaussian denoising and joint signal–interference reconstruction with frozen neural weights. On independent confirmation, it improves PSNR by 0.750 dB over Gaussian denoising and 1.334 dB over joint sampling. It also uses 34.7% fewer predictor evaluations than joint sampling. Fading, synchronization errors, and vehicular task accuracy require further evaluation.

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